

TRENT UNIVERSITY

Peterborough, Ontario, Canada K9J 7B8

DEPARTMENT OF ECONOMICS

Tel: (705) 748-1331, Fax: (705) 748-1567, E-Mail: ECONOMICS@TRENTU.CA

June 15, 2001

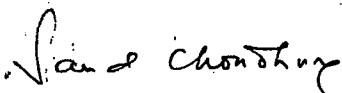
Mr. Michael Jack,
2699 Lakefield Road,
Peterborough, Ontario.
K9J 6X5

Dear Michael,

On behalf of the Department of Economics, I would like to congratulate you for winning the William B. Common Memorial Prize for the academic year 2000/2001. This prize is awarded annually to the student achieving the highest academic standing in first-year (Canadian) Macroeconomics.

It gives me great pleasure to present you with this book prize. Please accept the enclosed fifty dollar gift certificate from the Trent University Bookstore, along with our very best wishes for your future endeavours.

Yours sincerely,



Professor Saud Choudhry,
Chair.

cc: JBC Office



The Office of the Registrar at Trent University

June 18, 2002

Michael Jack
1049 Primrose Lane
Peterborough, ON
K9J 6X5

Dear Michael:

I am please to inform you that you have been awarded a Trent University Scholarship for the academic year 2002-2003 of the value of \$2000.

May I congratulate you on behalf of the Committee on scholarships, Awards, Bursaries and Financial Aid.

Sincerely,

Allan Barnfield
Assistant Registrar
Admissions and Recruitment



TRENT UNIVERSITY
FINANCIAL AID OFFICE

February 14, 2003

Michael Jack
Otonabee College

I am pleased to inform you that you have been awarded a bursary of \$300 from the **CITY OF PETERBOROUGH BURSARY FUND**.

The bursary has been credited to your student account first, and if applicable, a refund cheque is enclosed.

I hope this bursary will assist you in meeting your educational costs and to the successful completion of your year.

Yours truly,

Joyce Sutton
Financial Aid Manager



The Office of the Registrar at Trent University

June 12, 2003

Michael Jack
1049 Primrose Lane
Peterborough, On
K9J 6X5

Dear Michael:

I am pleased to inform you that you have been awarded a Trent University Scholarship in the value of \$2000 for the academic year 2003-2004. The scholarship will be automatically credited to your student account for September 2003.

May I congratulate you on behalf of the Committee on Admissions, Enrolment and Student Awards.

Sincerely,

Joyce Sutton
Financial Aid Manager



TRENT UNIVERSITY

Peterborough, Ontario, Canada K9J 7B8

Office of Research and Graduate Studies

December 11, 2003

Michael Jack
1049 Primrose Lane
Peterborough, ON., K9J 6X5

Dear Michael,

I am writing to inform you that your NSERC – PGS application was advanced to the national-level competition. On behalf of myself, the Graduate Sub-Committee on Scholarships (Science) and the Graduate Studies office, I would like to congratulate you on your successful thus far in the PGS competition. This year, the Graduate Sub-Committee on Scholarships was impressed with the number of very high quality applications. It is no small feat therefore to be selected as part of our quota of thirteen.

Please note that your file is now being reviewed by a national selection panel, the results of which will not be published until spring 2004. You should therefore not expect to hear from NSERC until mid to end April 2004.

Congratulations again, and good luck in the national PGS competition!

Sincerely,

Alison MacLeod
Scholarship Liaison Officer,
Research & Graduate Studies





Natural Sciences and Engineering
Research Council of Canada

Conseil de recherches en sciences
naturelles et en génie du Canada

350 Albert Street
Ottawa, Canada
K1A 1H5

350, rue Albert
Ottawa, Canada
K1A 1H5

March 17, 2004

Michael Jack
1049 PRIMROSE LANE
PETERBOROUGH ON K9J 6X5

Application No.: PGSM - 302027 - 2004
NSERC PIN: 282842

The Council wishes to inform you that your application for a scholarship was approved, based on your academic excellence, research potential, communication skills, and interpersonal and leadership abilities. Congratulations on your achievement!

You must inform NSERC at accept@nserc.ca of your decision to accept or decline the award by April 28, 2004; otherwise the offer will be cancelled. When corresponding with NSERC about this award, please indicate your Council application number shown above.

Please read the "Award Holder's Guide" carefully as it contains the regulations pertaining to your award, information about corresponding with NSERC, and forms for you to submit as required if you decide to take up your award. (This guide is available on our Web site at www.nserc.ca/guide/award/toc_e.htm.)

As a winner of an NSERC award, you may be eligible to apply for a supplement. You are encouraged to consult the "Program Guide for Students and Fellows" at www.nserc.gc.ca/sf_e.asp?nav=sfnv&lbi=2b for information about, and application procedures for, the different supplements.

CONDITIONS OF AWARD

- i) This award is subject to the conditions and regulations published by the Council in its scholarships and fellowships literature (on the Council's Web site) and to any amendments the Council may adopt.
- ii) Payments due in subsequent fiscal years are subject to availability of funds.

TYPE OF AWARD:	Postgraduate Scholarships - Master's (PGSM)
START DATE:	See the "Award Holder's Guide".
PLACE OF TENURE:	Your scholarship has been approved for tenure only at a Canadian university.
MAXIMUM VALUE AND DURATION OF AWARD:	\$17,300 over 12 months

Scholarships and fellowships are taxable (see the "Award Holder's Guide").

Teresa Brychcy
Director, Scholarships and Fellowships Programs

Canada

TRENT
UNIVERSITY
OFFICE OF THE REGISTRAR

June 18, 2004

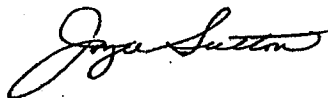
Mr. Michael Jack
1049 Primrose Lane
Peterborough, On
K9J 6X5

Dear Michael:

I am pleased to inform you that you have been awarded a scholarship from the EWING MEMORIAL SCHOLARSHIP FUND in the amount of \$2000 for the academic year of 2004-2005. This scholarship will be credited directly to your student account for use in September 2004.

May I congratulate you on behalf of the Committee on Admissions, Enrolment and Student Awards on your academic achievement.

Sincerely,



Joyce Sutton
Assistant Registrar
Financial Aid & Services



FINANCIAL AID OFFICE

May 18, 2004

Michael Jack
1049 Primrose Lane
Peterborough, ON
K9J 6X5

Dear Michael:

I am pleased to inform you that you have been awarded a *Summer Session Tuition Bursary* in the amount of \$562.

A bursary cheque is enclosed. If you have any outstanding fees due for summer session, please direct your payments to the Finance Office.

I hope this bursary assistance will enable you to complete your summer session course(s) successfully.

Yours sincerely,

A handwritten signature in cursive script, appearing to read "Joyce Sutton".

Joyce Sutton
Assistant Registrar, Financial Aid & Services

Js:cb
Encl.

Evaluation of Project

April 21, 2004

Good day Professors Hircock, and Feng,

I am pleased to submit a report on the software design project successfully completed this past academic year for the Department of Athletics and Recreation by a team lead by Michael Jack. I must first apologize for not being in attendance when they presented the programme to you and their peers. Unfortunately my time is often not mine to control and I was in Toronto at an Ontario University Athletics board meeting representing Trent.

In regards to the evaluation of the project, I have attempted to provide comments to substantiate a mark for Olga, David and Michael.

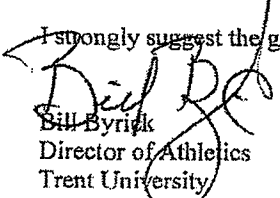
Introduction: The group met with me early last summer to discuss the opportunity and to scope out the needs of the Department. Through a series of meetings we identified the need for a computer based facility booking system, attendance monitoring system, statistical analysis need, as well as other requirements. After reviewing the manual system for booking we decided to embark on a project that would focus on satisfying this one need.

Planning: In preparation for the programme, we agreed to meet every Tuesday afternoon as required. It was made clear that there would be a time requirement on my part and those within the department to assist. Through a series of meetings the group gleaned significant information from the staff and added their creative suggestions to improve the procedure. Throughout the period the group continued to bring questions and possible solutions to our Tuesday meetings. Each week a prototype project began to evolve, and each week we were asked to confirm progress. The focus of the project was to produce a programme that would be more efficient, improve communication, provide more information to the student staff on the front line, and as a result provide more information to the customer. As frequent users of the facility this group understands the challenges and opportunities.

Result: I am very pleased with the result. I believe we have addressed the needs as outlined in the Planning Phase and the group created a system that will be both efficient and user friendly - a major need. Throughout the project the group added features that enhanced the product including the varying degrees of access, provision of statistical information, provision of visual directions for building setups, creation of daily and weekly schedules, etc.

Overall Comments: The group was very focused, professional and dedicated. They were also flexible in making changes and then adapting to the changes as time and needs arose. The department staff members have many priorities and as we are a customer service and deadline driven organization the group was forced to adapt to the frequent rescheduling of meetings. The group did adapt to these realities. We are pleased with the product. In regards to constructive criticism, the group was demanding and the project did take a significant amount of time. The reality is, the time was necessary in determining the need and finding a workable solution.

I strongly suggest the group receive an A+, or a 9.5 out of 10 for the project.


Bill Byrick
Director of Athletics
Trent University

MAY 5/04

billbyrick@trentu.ca
(705) 748-1252

TRENT UNIVERSITY

The Senate of the University on the recommendation
of the Faculty of Arts and Science
has conferred upon

Michael Jack

the degree of

Bachelor of Science in the Honours Program

Computer Science

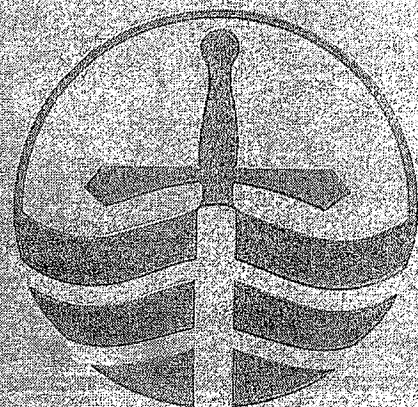
with all its rights, privileges
and obligations

Bonnie M. Patterson
President & Vice-Chancellor

Aileen Schibung
Registrar

Christine McKinnon
Dean of Arts & Science

Sam C. Storey
Head of College



Given at Peterborough, Ontario, Canada this 2nd day of June 2005

Official Transcript of Academic Record

Office of the Registrar, Blackburn Hall 1600 West Bank Drive, Peterborough, ON Canada K9J 7B8
Phone (705) 748-1215 Fax (705) 748-1629
E-mail registrar@trentu.ca Web www.trentu.ca



Susan Salusbury
SUSAN SALUSBURY
Registrar

Transcript valid only if bearing official signature and seal

To: **
Michael Jack
Official Transcript

Student Number: 0128039
Date of Birth: 1972 Dec 16
Issued On: 2006 Oct 16

Name: Michael Jack

Page: 1

Undergraduate

2000-2001 Academic Year

Basis of Admission: TRANSFER-TECH INST (FOREIGN)

		Credits	Mark	Grade
Computer Science	102H: Software Engin. With C/C++	0.5	94	A+
Economics	101H: Introductory Microeconomics	0.5	91	A+
Economics	102H: Introductory Macroeconomics	0.5	97	A+

THE WILLIAM B. COMMON, Q.C., LL.D. MEMORIAL PRIZE

2001 Summer Session

		Credits	Mark	Grade
Computer Science	104H: Programming Methodologies	0.5	97	A+

2001-2002 Academic Year

		Credits	Mark	Grade
Computer Science	230: Computer Organization	1.0	99	A+
Mathematics	110: Calculus of One Variable	1.0	94	A+
Computer Science	202H: Data Struct. & Algorithm Des.	0.5	82	A-
Mathematics	135H: Linear Algebra I: Matrix	0.5	83	A-
Computer Science	204H: Object-Oriented Programming	0.5	97	A+
Mathematics	155H: Introduction to Probability	0.5	90	A+
Mathematics	235H: Linear Algebra II: Vectors	0.5	90	A+

DEAN'S HONOURS ROLL

THE TRENT UNIVERSITY SCHOLARSHIP

2002-2003 Academic Year

		Credits	Mark	Grade
Mathematics	200: Calculus in Several Dimensions	1.0	90	AG
Mathematics	260: Discrete Structures	1.0	90	A+
Computer Science	332H: Fund. of Operating Systems	0.5	98	A+
Computer Science	362H: Object-Oriented Analysis & Des	0.5	87	A
Computer Studies	350H: Artificial Intelligence/Life	0.5	98	A+
Computer Science	304H: Software Specific: & Analysis	0.5	88	A
Computer Science	340H: Database Management Systems	0.5	97	A+
Computer Science	358H: Visual Programming Techniques	0.5	94	AG

DEAN'S HONOURS ROLL

Continued on Page 2

Official Transcript of Academic Record

Office of the Registrar, Blackburn Hall 1600 West Bank Drive, Peterborough, ON Canada K9J 7B8
Phone (705) 748-1215 Fax (705) 748-1629
E-mail registrar@trentu.ca Web www.trentu.ca



Susan Salusbury
SUSAN SALUSBURY
Registrar

Transcript valid only if bearing official signature and seal

To: **
Michael Jack
Official Transcript

Student Number: 0128039
Date of Birth : 1972 Dec 16
Issued On : 2006 Oct 16

Name: Michael Jack

Page: 2

Undergraduate

2003-2004 Academic Year

		Credits	Mark	Grade
Computer Science	337H: Systems Programming	0.5	97	A+
Computer Science	347H: Modelling and Simulation	0.5	89	A
Computer Science	401H: Software Engineering	0.5	91	A+
Computer Science	431H: Computer Networks	0.5	89	A
Computer Studies	261H: Info Systems for Bus & Gov't	0.5	87	A
Computer Science	305H: Theory of Computation	0.5	88	A
Computer Science	364H: Data Processing & File Struct.	0.5	99	A+
Computer Science	402: Software Engineering Project	1.0	96	A+
Computer Science	432H: Distributed Systems	0.5	90	A+

DEAN'S HONOURS ROLL
THE EWING MEMORIAL SCHOLARSHIP

2004 Summer Session

		Credits	Mark	Grade
Administrative Studies	102: Fundamentals of Accounting	1.0	91	A+
Computer Studies	345H: Societal Impact of Computers	0.5	93	A+
Computer Studies	396H: Human Computer Interaction	0.5	80	A-
Computer Studies	397H: Online Business Principles	0.5	97	A+
Psychology	101: Introduction to Psychology	1.0	77	B+

PRESIDENT'S HONOURS ROLL
DEAN'S HONOURS ROLL

Current Academic Status: Good Standing
Degree Completed : BACHELOR OF SCIENCE (HONOURS)
Majors : Computer Science
Minors :
Date : 02 Jun 2005

* End of Record *

מדינת ישראל
משרד החינוך
הוועדה להערכת תארים אקדמיים מחו"ל

אישור



אנו מאשרים בזה כי תעודה מס' *****

שניתנה

למרגב' ז' אק	מיכאל
מספר זיהוי 303933998	בתאריך 02/06/2005
ע"י טרנט יוניברסיטי	
בעיר אונטריו	בארץ קנדה
בתחום התמחות מדעי המחשב	

שקילה לתואר ראשון מבין התארים האקדמיים הנהוגים בישראל.

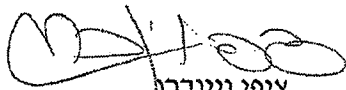
אישור זה ניתן לצורך דירוג בשכר בלבד.

האישור מקנה למחזיק/ה בו זכויות והטבות שכר הניתנות לבעלי תואר אקדמי כנ"ל.

הערכה זו אינה גורעת מזכויותיו של מוסד אקדמי בישראל, עפ"י חוק המועצה להשכלה גבוהה, תשי"ח-1958, בכל הנוגע לקבלת תלמידים ולהערכת תעודותיהם.

הערכה זאת אינה גורעת גם מזכויות הרשויות המוסמכות במדינה, בכל הנוגע להענקת רשיון לעסוק במקצוע.

אין באישור זה כדי לגרוע מזכותו של המעסיק בשירות המדינה או בשירות ציבורי אחר המבקש להיעזר בהערכה זו כדי לקבוע הוראות וכללים נוספים במסגרת שיקול דעתו לעניין דירוג ושכר.


ציפי ויינברג
ממונה

ירושלים, 06/01/2011
מס' האישור 339854

(הערכת תארים אקדמיים מחו"ל)

לכבוד,
ז'אק מיכאל
קק"ל 53/3
אשדוד 77471

TRENT UNIVERSITY

The Senate of the University on the recommendation
of the Committee on Graduate Studies
has conferred upon

Michael Jack

the degree of

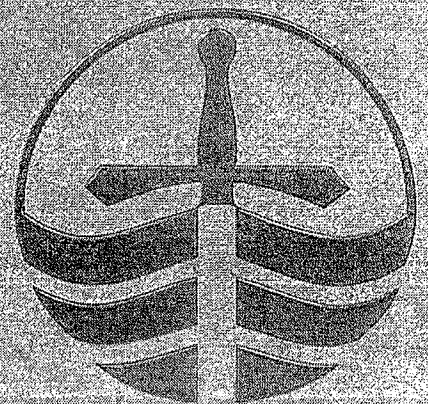
Master of Science

Applications of Modelling in the Natural & Social Sciences

with all its rights, privileges
and obligations

Bonnie M. Patterson
President & Vice-Chancellor

Aileen A. Leckie
Registrar



Angela Evans
Dean of Graduate Studies

Given at Peterborough, Ontario, Canada this 26th day of September 2006



TRENT UNIVERSITY, PETERBOROUGH, ONTARIO CANADA
Graduate Academic Record Card

Name Jack, Michael Student ID 128039 (6056001192)
Home Address 1049 Primrose Lane Date of Birth December 16, 1972
Peterborough, ON K9J 6X5 Citizenship Canadian
Present Address _____ Date of Enrolment September 1, 2004
Earlier Degree(s) H.B.Sc. (Trent) Basis of Admission Honours B.Sc.
Program Modelling in the Natural & Social Sciences

PROGRAM OF STUDIES LEADING TO THE DEGREE OF Master of Science

Approved Courses							
Subject	No.	Session	Final Grade	Subject	No.	Session	Final Grade
Applications of Modelling	AMOD 501H	2004FW	A+ (93%)	Mathematical Aspects of Modelling	AMOD 571H	2005WI	A (85%)
Classic Artificial Intelligence	AMOD 502H	2004FW	A (85%)	Seminar on Applications of Modelling	AMOD 590	2005FW	Pass
The Foundations of Modelling	AMOD 561H	2004FW	A (88%)				

Thesis Workload Modeling and Internal Backfilling for Parallel Job Scheduling

Supervisor B. Patrick

Members of Supervisory Committee R. Hurley, & B. Atkinson,

Date Degree Conferred October 2006 (Degree Requirement Met: September 2006)

Awards NSERC Post Graduate Scholarship - Masters 2005-06

SIGNED: _____

DATE: Oct 26/06

This record is not an official transcript unless signed by the Dean of Graduate Studies and stamped with the University seal.

**WORKLOAD MODELING AND INTERNAL
BACKFILLING FOR PARALLEL JOB SCHEDULING**

MICHAEL JACK

2006

Workload Modeling and Internal Backfilling for Parallel Job Scheduling

A Thesis Submitted to the Committee on Graduate Studies
in Partial Fulfillment of the Requirements for the Degree of Master of Science
in the Faculty of Arts and Science

TRENT UNIVERSITY
Peterborough, Ontario, Canada

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Applications of Modelling in the Natural and Social Sciences

M.Sc. Program

October 2006

ABSTRACT

Workload Modeling and Internal Backfilling for Parallel Job Scheduling

Michael Jack

Parallel job scheduling for high-performance computing has been concentrated in two distinct research areas. First, real job schedulers have been developed, tested, and tweaked in response to real workload demands. Second, theoretical models of both job schedulers and workloads have been developed to generate theoretical results for selected metrics. In this thesis, we strive to bridge the gap between practice and theory, and to develop theoretical models that are grounded on real job schedulers and workload data. First, we gather, filter, standardize, and analyze real workload data from three High-Performance Computing centers. Metrics are evaluated for average wait time, average runtime, and average slowdown. Second, we develop a job scheduler model using a First-Come, First-Served strategy with and without EASY backfilling. The workload for which the job scheduler model best approximates the actual metrics is selected for further investigation. Third, empirical distributions are derived for inter-arrival time, job size, and job length in order to create synthetic workloads. Metrics are averaged over several runs using the job scheduler model on the synthetic workloads and again compared to the actual metrics for the real system. Finally, the job scheduler and workload models are extended to incorporate internal backfilling as well. It is shown that internal backfilling lowers average wait time and slowdown, improves system utilization, and therefore, offers better throughput.

Keywords: workload modeling, parallel job scheduling, backfilling, internal fragmentation.

מדינת ישראל
משרד החינוך
הוועדה להערכת תארים אקדמיים מחו"ל



אישור

אנו מאשרים בזה כי תעודה מס' *****

שניתנה

למ/גב' ז' אק	מיכאל
מספר זיהוי 303933998	בתאריך 25/09/2006
ע"י טרנט יוניברסיטי	
בעיר אונטריו	בארץ קנדה
בתחום התמחות	
ישומים של מודלים במדעי הטבע והחברה	

שקילה לתואר שני מבין התארים האקדמיים הנהוגים בישראל.

אישור זה ניתן לצורך דירוג בשכר בלבד.

האישור מקנה למתזיק/ה בו זכויות והטבות שכר הניתנות לבעלי תואר אקדמי כנ"ל.

הערכה זו אינה גורעת מזכויותיו של מוסד אקדמי בישראל, עפ"י חוק המועצה להשכלה גבוהה, תשי"ח-1958, בכל הנוגע לקבלת תלמידים ולהערכת תעודותיהם.

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ציפי ויינברג
ממונה

(הערכת תארים אקדמיים מחו"ל)

ירושלים, 06/01/2011
מס' האישור 339858

לכבוד,
ז' אק. מיכאל
קק"ל 53/3
אשדוד 77471



Trent University
Computer Science 230H
2007 Fall Term

COMPUTER ORGANIZATION

Course Outline

Instructor: Michael Jack, *E-mail: michaeljack@trentu.ca*
Office: OC 102, Conference Room
Office hours: Monday, Wednesday, 15:00 - 17:00.

Secretary: Bonnie MacKinnon, *E-mail: compstudies@trentu.ca*
Office: OC 102.6, Phone: (705) 748-1011 ext.1495

Website: <http://www.trentu.ca/webct>

Texts:

Required:

Patterson, D. and Hennessy, J. Computer Organization and Design, 3rd edition, 2005.
Jack, M. Computer Science 230H Lecture Notes.
Kjell, B. Programmed Introduction to MIPS Assembly Language
(<http://programmedlessons.org/AssemblyTutorial>)

Supplementary:

Stallings, W., Computer Organization and Architecture: Designing for Performance 7th edition, 2006.

Course Description:

Organization of a simple stored-program computer: CPU, busses and memory. Instruction sets, machine code, and assembly language. Conventions for assembly language generated by compilers. Floating-point number representation. Hardware organization of simple processors. Address translation and virtual memory. Very introductory examples of interrupt handling, multi-tasking systems, storage systems, and interconnections.

Course Objectives:

- Basic understanding of computer organization: roles of processors, main memory, and secondary storage systems.
- Understanding the concept of pipelining and why it is needed.
- Understanding arithmetic and logical operations with integer operands.
- Understanding floating-point number systems and operations.
- Understanding simple data path and control designs for processors.
- Understanding memory organization, including cache structures and virtual memory schemes.
- Understanding the concept of programs as sequences of machine instructions.
- Understanding the relationship between assembly language and machine language.
- Understanding the relationship between high-level compiled languages and assembly language.
- Development of skill in assembly language programming.

Course content:

Lectures:

- Introduction
- Instruction Set Architecture (ISA)
- Pipelining
- Computer Arithmetic - ALU
- Control Unit - CPU
- Memory Hierarchy
- Storage Systems

Seminars:

MIPS32 assembly language programming using SPIM simulator.

Class meetings:

Lectures are Mondays 17:00 - 19:00 in GCS 110.
Seminars are Wednesdays 17:00 - 19:00 in GCS 110.

Grading scheme:

6%	6 topic summaries
30%	6 quizzes
30%	3 assembler language programming assignments
34%	Final exam

Notes:

For each of the lecture topics you are to prepare a brief summary of what you think the topic is about. The summary should be limited to a page. Use any resources such as Internet and books and provide reference for the source from where you acquired the information. The summary is to be submitted before the lecture on the respective topic begins. For example, on Monday, September 17th, you are to hand in a summary on Instruction Set Architecture (ISA). Failure to submit a summary before the class will result in assignment of grade 0 for the respective summary.

Examination policy:

All quizzes are open text book. The duration of each quiz is one hour. Final exam is open text book. The duration of the final exam is three hours. The final exam will be structured around quiz questions and assembly language programming assignments. The better you do on the quizzes and assignments, the easier it will be for you to "ace" the final.

No computers may be used during the quizzes and the final examination. The only digital devices permitted are calculators and digital wristwatches. Cell phones must be turned off during examinations.

Assignment submission policy:

All assignments are to be submitted using the myLearningSystem (WebCT) assignment drop box tool. Do NOT email the assignments or slide them under anyone's door. They will NOT be accepted. For each assignment there is a due date and a cutoff date. Any assignment submitted past the due date will incur a penalty of 10% per day. No assignments will be accepted past the cutoff date.

Missed or Delayed Term Work:

You are expected to complete all programming assignments and hand them in on time, and you are expected to write the quizzes at the scheduled time.

If you don't hand in some term work due to illness or another legitimate reason, Michael Jack is willing to make accommodations, but only if you report the problem to Michael Jack (not anyone else) as soon as is reasonably possible. Accommodations for missed term work will be decided on a case-by-case basis; typically they will involve either accepting assignments after the due date or making changes to the weighting used to compute the course grade.

Academic Dishonesty:

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure in an assignment to suspension from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's Academic Dishonesty Policy which is printed in the University Calendar.

Get help, but do not cheat!

You are encouraged to discuss the assignments with the instructor, lab adviser, and fellow students, since this is one of the best ways to learn the material. However, you should not let anyone write your programs or solve other exercises for you. When you hand in your assignments, ask yourself two questions:

*Do I understand all the material I am handing in?
Could I do this assignment over again without any help?*

The answer to both questions should be YES.

Computers allow electronic copying of programs, which makes it very easy to cheat in a course like COSC 230. If you are caught cheating you will be reported to the Dean's Office for appropriate discipline. If you cheat and don't get caught, you are still in trouble, because examination marks count more than assignment marks in your final course grade.

Access to Instruction:

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Disability Services Office (BL Suite 109, 748-1281, disabilityservices@trentu.ca) as soon as possible. Complete text can be found under Access to Instruction in the Academic Calendar.

Proposed Course Schedule (may slightly change as the course progresses):

Date	Type	Location	Start Time	Duration	Topic
September 10	Lecture 1	GCS 110	17:00	2 hrs.	Introduction
September 12	Seminar 1	GCS 110	17:00	2 hrs.	MIPS
September 17	Lecture 2	GCS 110	17:00	2 hrs.	Instruction Set Architecture
September 19	Seminar 2	GCS 110	17:00	2 hrs.	MIPS
September 24	Lecture 3	GCS 110	17:00	2 hrs.	Instruction Set Architecture
September 26	Quiz 1 and Seminar 3	GCS 110	17:00	2 hrs.	Instruction Set Architecture, MIPS
October 1	Lecture 4	GCS 110	17:00	2 hrs.	Pipelining
October 3	Seminar 4	GCS 110	17:00	2 hrs.	MIPS
October 8 Thanksgiving	-	-	-	-	-
October 10	Quiz 2 and Lecture 5	GCS 110	17:00	2 hrs.	Pipelining, Computer Arithmetic
October 15	Lecture 6	GCS 110	17:00	2 hrs.	Computer Arithmetic
October 17	Seminar 5	GCS 110	17:00	2 hrs.	MIPS
October 22 and 24 Reading week	-	-	-	-	-
October 29	Lecture 7	GCS 110	17:00	2 hrs.	Computer Arithmetic
October 31	Quiz 3 and Seminar 6	GCS 110	17:00	2 hrs.	Computer Arithmetic, MIPS
November 5	Lecture 8	GCS 110	17:00	2 hrs.	Computer Arithmetic
November 7	Quiz 4 and Seminar 7	GCS 110	17:00	2 hrs.	Computer Arithmetic, MIPS
November 12	Lecture 9	GCS 110	17:00	2 hrs.	CPU Control Unit
November 14	Quiz 5 and Seminar 8	GCS 110	17:00	2 hrs.	CPU Control Unit, MIPS
November 19	Lecture 10	GCS 110	17:00	2 hrs.	Memory Hierarchy
November 21	Seminar 9	GCS 110	17:00	2 hrs.	MIPS
November 26	Lecture 11	GCS 110	17:00	2 hrs.	Memory Hierarchy
November 28	Quiz 6 and Seminar 10	GCS 110	17:00	2 hrs.	Memory Hierarchy, MIPS
December 3	Lecture 12	GCS 110	17:00	2 hrs.	Storage Systems
December 5	Seminar 11	GCS 110	17:00	2 hrs.	Final exam review



Trent University
Computer Science 204H
2007 Fall Term

OBJECT-ORIENTED PROGRAMMING

Course Outline

Instructor: Michael Jack, *E-mail: michaeljack@trentu.ca*
Office: OC 102, Conference Room
Office hours: Monday, Wednesday, 15:00 - 17:00.

Secretary: Bonnie MacKinnon, *E-mail: compstudies@trentu.ca*
Office: OC 102.6, Phone: (705) 748-1011 ext.1495

Website: <http://www.trentu.ca/webct>

Texts:

Required:

Lewis, J. C# Software Solutions, 1st edition, 2007.
Jack, M. Computer Science 204H Lecture Notes.

Supplementary:

Deitel Developer Series. C# for Programmers, 2nd edition, 2006.
Deitel. C# How to Program, 1st edition, 2002.

Course Description:

The purpose of this course is to continue developing and enhancing the concepts of object-orientation and learn the key principles of object-oriented programming using C# programming language.

Course Objectives:

- Introducing C# programming language and the concepts of object-oriented programming.
- Understanding the basic types of data used in C# program and the use of expressions to perform calculations.
- Understanding Boolean expressions and how to use them to control the flow of programs.
- Understanding the use of predefined classes in the .NET architecture and the objects that can be created from them.
- Understanding the basic issues related to writing classes and methods.
- Understanding arrays and array processing.
- Developing the techniques for identifying classes and objects needed for a problem and the relationships among them.
- Understanding class derivations and associated concepts such as class hierarchies, overriding, and visibility.
- Understanding the concept of binding and how it relates to polymorphism.
- Understanding exceptions.

Course content:**Lectures:**

- Introduction
- Data and Expressions
- Conditionals and Loops
- Using Classes and Objects
- Writing Classes
- Arrays
- Object-Oriented Design
- Inheritance
- Polymorphism
- Exceptions

Seminars:

Discussing programming examples and assignments as they relate to the lecture topics.

Meetings:

Lectures are Tuesday 17:00 - 19:00 in GCS 112.
Seminars are Tuesday 19:00 - 20:00 in GCS 112.

Grading scheme:

20%	9 online quizzes with 8 best selected for the final grade
40%	8 programming assignments
40%	Final exam

Notes:

For each of the lecture topics you have an online quiz on the myLearningSystem (WebCT) and a programming assignment.

Examination policy:

Since you will be taking online quizzes at the time of your convenience you can use any resources such as Internet and books to help you answer the questions. The duration of each quiz varies from half-an-hour to an hour depending on the scope and complexity of the quiz.

Final exam is open text book. The duration of the final exam is three hours. The final exam will be structured around quiz questions and C# language programming assignments. The better you do on the quizzes and assignments, the easier it will be for you to "ace" the final.

No computers may be used during the final examination. The only digital devices permitted are calculators and digital wristwatches. Cell phones must be turned off during the examination.

Assignment submission policy:

All assignments are to be submitted using the myLearningSystem (WebCT) assignment drop box tool. Do NOT email the assignments or slide them under anyone's door. They will NOT be accepted.

For each assignment there is a due date and a cutoff date. Any assignment submitted past the due date will incur a penalty of 10% per day. No assignments will be accepted past the cutoff date.

Missed or Delayed Term Work:

You are expected to complete all programming assignments and hand them in on time, and you are expected to write the quizzes at the scheduled time.

If you don't hand in some term work due to illness or another legitimate reason, Michael Jack is willing to make accommodations, but only if you report the problem to Michael Jack (not anyone else) as soon as is reasonably possible. Accommodations for missed term work will be decided on a case-by-case basis; typically they will involve either accepting assignments after the due date or making changes to the weighting used to compute the course grade.

Academic Dishonesty:

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure in an assignment to suspension from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's Academic Dishonesty Policy which is printed in the University Calendar.

Get help, but do not cheat!

You are encouraged to discuss the assignments with the instructor, lab adviser, and fellow students, since this is one of the best ways to learn the material. However, you should not let anyone write your programs or solve other exercises for you. When you hand in your assignments, ask yourself two questions:

*Do I understand all the material I am handing in?
Could I do this assignment over again without any help?*

The answer to both questions should be YES.

Computers allow electronic copying of programs, which makes it very easy to cheat in a course like COSC 204. If you are caught cheating you will be reported to the Dean's Office for appropriate discipline. If you cheat and don't get caught, you are still in trouble, because examination marks count more than assignments marks in your final course grade.

Access to Instruction:

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Disability Services Office (BL Suite 109, 748-1281, disabilityservices@trentu.ca) as soon as possible. Complete text can be found under Access to Instruction in the Academic Calendar.

Course Schedule (may slightly change as the course progresses):

Date	Type	Start Time	Location	Duration	Topic	Due back
September 11	Lecture 1	16:00	GCS 112	3 hrs.	Introduction	
September 18	Lecture 2	16:00	GCS 112	3 hrs.	Data and Expressions	Assign 1
September 25	Lecture 3	16:00	GCS 112	3 hrs.	Using Classes and Objects	Assign 2
October 2	Lecture 4	16:00	GCS 112	3 hrs.	Writing Classes	Assign 3
October 9	Lecture 5	16:00	GCS 112	3 hrs.	(no lecture; reading only)	-
October 16	Lecture 6	16:00	GCS 112	3 hrs.	Arrays	-
October 23 Reading week	-	-	-	-	-	-
October 30	Lecture 7	16:00	GCS 112	3 hrs.	Object-Oriented Design	Assign 4
November 6	Lecture 8	16:00	GCS 112	3 hrs.	Inheritance	Assign 5
November 13	Lecture 9	16:00	GCS 112	3 hrs.	(no lecture; reading only)	-
November 20	Lecture 10	16:00	GCS 112	3 hrs.	Polymorphism	Assign 6
November 27	Lecture 11	16:00	GCS 112	3 hrs.	Exceptions	Assign 7
December 4	Lecture 12	16:00	GCS 112	3 hrs.	Final exam review	Assign 8



Trent University
ADMIN-COIS 1620H
2008 Summer Term

INTRO TO INFORMATION SYSTEMS

Course Outline

Instructor: Michael Jack,
OC 102
(705) 748-1011 ext. 7251
michaeljack@trentu.ca

Secretary: Bonnie MacKinnon
OC 102.6
(705) 748-1011 ext. 7802
cois@trentu.ca

Text: *Information Systems Today: Why IS Matters (2nd Canadian Edition)*,
L.M. Jessup, J.S. Valacich, and M.R. Wade (2008),
Pearson Prentice Hall (www.pearsoned.ca/jessup)

Meetings: Tuesday/Thursday, 18:30 - 20:30, GCS 110.

Office Hours: Monday, 17:00 - 18:30, OC 102.

Course Content:

This course provides an introductory overview of information systems and will address the following important questions:

- Why Information Systems Matter
- Information Systems for Competitive Advantage
- Data and Knowledge Management
- The Internet and Security
- Electronic Business, Intranets, and Extranets
- Organizational Information Systems
- Enterprise-Wide Information Systems
- Information Systems Development and Acquisition
- Information Systems Ethics and Computer Crime

Course Objectives:

- Understand why information systems matter to contemporary organizations and society.
- Understand exactly what innovative organizations are doing with contemporary information systems and where things are headed.
- Present a uniquely Canadian perspective on the field of information systems.
- Empower students with the essential knowledge they need to be successful in the use and understanding of information technology in their careers.
- Teach the students the principles and importance of critical thinking.

Methodology:

For each student, this course is comprised of weekly lectures, nine quizzes, two tests, and a research project. It is a busy agenda, but one that attempts to involve you as much as possible in the learning process.

Grading scheme:

40%	9 online quizzes
40%	2 in-class tests
20%	1 research project
5%	bonus

Quizzes:

The quizzes are based on the chapters from the text book – one quiz per chapter. The best 8 out of 9 will be selected for the calculation of the final grade. All the quizzes will be done online using the myLearningSystem (WebCT) system. Since you are taking quizzes at the time and place of your convenience (during the allocated time frame), you can use any resources such as the Internet and books to help you answer the questions. However, the quizzes must be completed independently. The duration of each quiz will be 30 minutes.

Tests:

Both tests are paper based and will be given in class settings. The tests are open text book. The duration of each test is one hour. The tests are structured strictly around material presented in each chapter. The better you do on your quizzes, the easier it will be for you to “ace” the tests.

No computers may be used during the tests. The only digital devices permitted are calculators and digital wristwatches. Cell phones must be turned off as well.

Research project:

The intent of the research project is to teach you to think critically and to look at Information Systems from a broader perspective. You are to pick a case in which Information Systems inadvertently or intentionally produce adverse effects on the environment, animals, or humans. You are required to write an 8-10 page summary of what you have found and what you think of it. Make sure to provide references to the sources of your information. The project paper is to be completed independently and submitted in paper form on the last day of the classes.

A list of possible topics to look into:

- www.consciousmedianetwork.com
 - Nick Begich on Chipping
 - Nick Begich on Cellphone Radiation
 - Nick Begich on the Human RFID chip
 - Ian Wagge on our Technological Future
 - Leuren Moret on The Uranium Wars
 - Jerry E. Smith on HAARP

- <http://www.storyofstuff.com>
- Zeitgeist (available on video.google.com)
- America Freedom to Fascism (available on video.google.com)

Bonus:

You can earn up to 5% bonus that will be added to your final grade. For every logical or syntax mistake that you find in the text book you can earn a point. The point will be awarded to ONLY the first student who shows the instructor the mistake and for a maximum of 5 points per student.

Missed or Delayed Term Work:

You are expected to complete your research projects and hand them in on time and you are expected to take the online quizzes within the allocated time frame. If you miss or don't hand in some term work due to illness or another legitimate reason, Michael Jack is willing to make accommodations, but only if you report the problem to Michael Jack (not anyone else) as soon as is reasonably possible. Accommodations for missed term work will be decided on a case-by-case basis. Typically they will involve either taking the quiz after the expiration date or making changes to the weighting used to compute the course grade.

Correspondence:

Due to privacy legislation, any email correspondence from the course instructor or assistant will be forwarded to your Trent account only.

Academic Dishonesty:

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure in an assignment to suspension from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's Academic Dishonesty Policy which is printed in the University Calendar and found at:

http://www.trentu.ca/deansoffice/policies_dishonesty.php

Get help, but do not cheat!

You are encouraged to discuss your work with the instructor, lab adviser, and fellow students, since this is one of the best ways to learn the material. However, you should not let anyone take your quizzes and do your research project for you. When you submit your work, ask yourself two questions:

*Do I understand all the material I am handing in?
Could I do this over again without any help?*

The answer to both questions should be YES. If you are caught cheating you will be reported to the Dean's Office for appropriate discipline.

Access to Instruction:

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Disability Services Office (BL Suite 109, 748-1281, disabilityservices@trentu.ca) as soon as possible. Complete text can be found under Access to Instruction in the Academic Calendar.

Course Schedule

Timeframe		Material
Release	Close	
June 17	June 22	Ch 1: Why Information Systems Matter.
June 22	June 25	Ch 2: Information Systems for Competitive Advantage.
June 25	June 29	Ch 3: Data and Knowledge Management.
June 29	July 2	Ch 4: The Internet and Security.
July 3	July 3	Test 1: Ch 1, 2, 3, 4.
July 2	July 8	Ch 5: Electronic Business, Intranets, and Extranets.
July 8	July 13	Ch 6: Organizational Information Systems.
July 13	July 18	Ch 7: Enterprise-Wide Information Systems.
July 18	July 23	Ch 8: Information Systems Development and Acquisition.
July 23	July 28	Ch 9: Information Systems Ethics and Computer Crime.
July 29	July 29	Test 2: Ch 5, 6, 7, 8, 9; Research project summary.



Trent University
COIS 1020H
2008 Summer Term

INTRODUCTION TO COMPUTING WITH C#

Course Outline

Instructor: Michael Jack,
OC 102
(705) 748-1011 ext. 7251
michaeljack@trentu.ca

Secretary: Bonnie MacKinnon
OC 102.6
(705) 748-1011 ext. 7802
cois@trentu.ca

Text:

Required:

Chegwidden, J. and Gaddis, T. Starting Out with C#.
Accompanying Slides by Scott Jones Publishing.

Supplementary:

Deitel. C# How to Program, 1st edition, 2002.
Arthur Gittleman, Computing with C# and the .NET Framework, 2003

Prize:

Deitel Developer Series, C# for Programmers, 2nd edition, 2006 will be awarded to the student who has achieved the highest academic standing in the course.
Lewis, J. C# Software Solutions, 1st edition, 2007 will be awarded to the student who has achieved the highest cumulative grade for programming assignments in the course.

Meetings: Monday/Wednesday, 18:30 - 20:30, OC 203.

Office Hours: Monday, 17:00 - 18:30, OC 102.

Grading scheme:

9%	3 labs
35%	7 assignments
28%	8 quizzes
28%	final exam
5%	bonus

Course Content:

The purpose of this course is to examine the fundamental issues of software development including algorithmic design, implementation, testing, and documentation. The course introduces the basic constructs of sequence, selection, repetition, data representation, classes, arrays, and methods in the context of object-oriented technologies.

- Introduction
- Data Types
- Operators and Expressions
- Methods
- Classes and Objects
- Conditionals
- Files
- Repetition constructs
- Arrays
- More on Classes and Objects

Examination policy:

The quizzes are based on the chapters from the text book – one quiz per chapter. The best 7 out of 8 will be selected for the calculation of the final grade. All the quizzes will be done online using the myLearningSystem (WebCT) system. Since you are taking quizzes at the time and place of your convenience (during the allocated time frame), you can use any resources such as the Internet and books to help you answer the questions. However, the quizzes must be completed independently. The duration of each quiz will be 20 minutes.

The programming assignments are based on the chapters from the text book. Each assignment includes two programming questions. You are to complete at least one question. If you complete both questions (which is recommended), the assignment grade will be the best mark of the two.

Final exam is open text book. The final exam will be based on the same material as the quizzes, labs, and programming assignments. The better you do during the course, the easier it will be for you to “ace” the final. The duration of the final exam will be 2 hours.

No computers may be used during the final exam. The only digital devices permitted are calculators and digital wristwatches. Cell phones must be turned off during the examination.

You can earn up to 5% bonus that will be added to your final grade. For every logical or syntax mistake that you find in the text book you can earn a point. The point will be awarded to ONLY the first student who shows the instructor the mistake and for a maximum of 5 points per student.

Assignment submission policy:

The programming assignments are to be submitted using the myLearningSystem (WebCT) assignment drop box tool. Do NOT email the assignments or print and slide them under anyone's door as they will NOT be accepted. For each assignment and lab there is a due date and a cutoff date. Any assignment or lab submitted past the due date will incur a penalty of 10% per day. No assignments and labs will be accepted past the cutoff date.

Missed or Delayed Term Work:

You are expected to complete all labs and programming assignments and hand them in on time, and you are expected to write the quizzes within the allocated time frame.

If you don't hand in some term work due to illness or another legitimate reason, Michael Jack is willing to make accommodations, but only if you report the problem to Michael Jack (not anyone else) as soon as is reasonably possible.

Accommodations for missed term work will be decided on a case-by-case basis; typically they will involve either accepting assignments after the due date or making changes to the weighting used to compute the course grade.

Academic Dishonesty:

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure in an assignment to suspension from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's Academic Dishonesty Policy which is printed in the University Calendar and found at:

http://www.trentu.ca/deansoffice/policies_dishonesty.php

Get help, but do not cheat!

You are encouraged to discuss your work with the instructor, lab adviser, and fellow students, since this is one of the best ways to learn the material. However, you should not let anyone take your quizzes, prepare your labs and program your assignments for you. When you submit your work, ask yourself two questions:

*Do I understand all the material I am handing in?
Could I do this over again without any help?*

The answer to both questions should be YES. Computers allow electronic copying of programs, which makes it very easy to cheat in a course like COIS 1020H. If you are caught cheating you will be reported to the Dean's Office for appropriate discipline!

Access to Instruction:

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Disability Services Office (BL Suite 109, 748-1281, disabilityservices@trentu.ca) as soon as possible. Complete text can be found under Access to Instruction in the Academic Calendar.

Proposed Course Schedule (may slightly change as the course progresses):

Date	Type	Location	Start Time	Duration	Topic
June 16	Lecture 1	OC 203	18:30	2 hrs.	Introduction
June 18	Lecture 2	OC 203	18:30	2 hrs.	Data Types
June 23	Lecture 3	OC 203	18:30	2 hrs.	Operators and Expressions
June 25	Lecture 4	OC 203	18:30	2 hrs.	Methods
June 30	-	-	-	-	-
July 2	Lecture 5	OC 203	18:30	2 hrs.	Classes and Objects
July 7	Lecture 6	OC 203	18:30	2 hrs.	Classes and Objects
July 9	Lecture 7	OC 203	18:30	2 hrs.	Conditionals
July 14	Lecture 8	OC 203	18:30	2 hrs.	Repetition constructs
July 16	Lecture 9	OC 203	18:30	2 hrs.	Arrays
July 21	Lecture 10	OC 203	18:30	2 hrs.	More on Classes and Objects
July 23	Lecture 11	OC 203	18:30	2 hrs.	Review
July 28	Test	OC 203	18:30	2 hrs.	Final Examination



Trent University
Computer Science 102H
2008 Winter Term

INTRODUCTION TO COMPUTING

Course Outline

Instructor: Michael Jack, *E-mail: michaeljack@trentu.ca*
Office: UL 10
Office hours: Thursday 21:10 - 22:00

Secretary: Bonnie MacKinnon, *E-mail: compstudies@trentu.ca*
Office: OC 102.6, Phone: (705) 748-1011 ext.1495

Website: <http://www.trentu.ca/webct>

Texts:

- **Required:**
Chegwidden, J. and Gaddis, T. Starting Out with C#. Accompanying Slides by Scott Jones Publishing.

Supplementary:

Deitel. C# How to Program, 1st edition, 2002.
Arthur Gittleman, Computing with C# and the .NET Framework, 2003
Deitel Developer Series. C# for Programmers, 2nd edition, 2006.

Course Description:

The purpose of this course is to examine the fundamental issues of software development including algorithmic design, implementation, testing, and documentation. The course introduces the basic constructs of selection, iteration, data representation, classes, and methods in the context of object-oriented technologies.

Course Content:

- Introduction
- Data Types
- Operators and Expressions
- Methods
- Classes and Objects
- Conditionals
- Files
- Repetition constructs
- Arrays
- Algorithmic Design
- Testing and Documentation
- Further examination of classes and objects

Meetings:

Thursday 19:10 - 21:00 in UL 10.

Grading scheme:

18%	3 programming labs
20%	2 programming assignments
42%	7 online quizzes using the myLearningSystem (WebCT) system (6 best are taken for the calculation of the final grade)
20%	Final exam

Examination policy:

Since you are taking online quizzes at the time of your convenience you can use any resources such as Internet and books to help you answer the questions. The duration of each quiz is 30 minutes.

Final exam is open text book. The duration of the final exam is two hours. The final exam is structured around quiz questions, labs, and programming assignments. The better you do during the course, the easier it will be for you to "ace" the final.

No computers may be used during the final examination. The only digital devices permitted are calculators and digital wristwatches. Cell phones must be turned off during the examination.

Assignment submission policy:

The assignments are to be submitted using the myLearningSystem (WebCT) assignment drop box tool. Do NOT email the assignments or slide them under anyone's door. They will NOT be accepted. For each assignment and lab there is a due date and a cutoff date. Any assignment or lab submitted past the due date will incur a penalty of 10% per day. No assignments and labs will be accepted past the cutoff date.

Missed or Delayed Term Work:

You are expected to complete all labs and programming assignments and hand them in on time, and you are expected to write the quizzes within the allocated time frame.

If you don't hand in some term work due to illness or another legitimate reason, Michael Jack is willing to make accommodations, but only if you report the problem to Michael Jack (not anyone else) as soon as is reasonably possible. Accommodations for missed term work will be decided on a case-by-case basis; typically they will involve either accepting assignments after the due date or making changes to the weighting used to compute the course grade.

Academic Dishonesty:

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure in an assignment to suspension from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's Academic Dishonesty Policy which is printed in the University Calendar.

Get help, but do not cheat!

You are encouraged to discuss the assignments with the instructor, lab adviser, and fellow students, since this is one of the best ways to learn the material. However, you should not let anyone write your programs or solve other exercises for you. When you hand in your assignments, ask yourself two questions:

*Do I understand all the material I am handing in?
Could I do this assignment over again without any help?*

The answer to both questions should be YES.

Computers allow electronic copying of programs, which makes it very easy to cheat in a course like COSC 102H. If you are caught cheating you will be reported to the Dean's Office for appropriate discipline. If you cheat and don't get caught, you are still in trouble, because examination marks count more than assignments marks in your final course grade.

Access to Instruction:

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Disability Services Office (BL Suite 109, 748-1281, disabilityservices@trentu.ca) as soon as possible. Complete text can be found under Access to Instruction in the Academic Calendar.



Trent University
Computer Science 232H
2008 Winter Term

DIGITAL LOGIC

Course Outline

Instructor: Michael Jack, E-mail: michaeljack@trentu.ca
Office: OC 102
Office hours: Monday, 14:00 - 16:00.

Secretary: Bonnie MacKinnon, E-mail: compstudies@trentu.ca
Office: OC 102.6, Phone: (705) 748-1011 ext.1495

Website: <http://www.trentu.ca/webct>

Texts: M. Moris. Mano and M. D. Ciletti, Digital Design (fourth edition)
Tessier, R. and Jack, M. Computer Science 232H Lecture Slides

Course Description:

The purpose of this course is to provide a solid foundation in Boolean algebra, truth tables, Karnaugh maps, minimization of circuits, adders, multiplexers, flip-flops, sequential logic design and memory circuits. The investigation of digital logic circuits is intended to provide a deeper understanding of the internal workings of microprocessors and memory chips.

Meetings: Lectures are Monday 16:00 - 18:00 in GCS 110.
Seminars are Monday 18:00 - 19:00 in GCS 110.

Course Schedule:

Date	Type	Start Time	Location	Duration	Due back
January 7	Lecture 1	17:00	GCS 110	2 hrs.	
January 14	Lecture 2	17:00	GCS 110	2 hrs.	
January 21	Lecture 3	17:00	GCS 110	2 hrs.	
January 28	Lecture 4	17:00	GCS 110	2 hrs.	
February 4	Lecture 5	17:00	GCS 110	2 hrs.	Assign 1
February 11	Lecture 6	17:00	GCS 110	2 hrs.	
Reading week	-	-	-	-	
February 25	Midterm	17:00	GCS 110	2 hrs.	Assign 2
March 3	Lecture 7	17:00	GCS 110	2 hrs.	
March 10	Lecture 8	17:00	GCS 110	2 hrs.	
March 17	Lecture 9	17:00	GCS 110	2 hrs.	
March 24	Lecture 10	17:00	GCS 110	2 hrs.	Assign 3
March 31	Lecture 11	17:00	GCS 110	2 hrs.	
TBD	Final	TBD	TBD	3 hrs.	Assign 4

Grading scheme:

40%	4 assignments
20%	Midterm
40%	Final exam

Examination policy:

Midterm and final exams are open text book. The duration of the midterm is 2 hours. The duration of the final exam is 3 hours. Both exams are structured around assignment questions. The better you do on the assignments, the easier it will be for you to "ace" the exams.

No computers may be used during the exams. The only digital devices permitted are calculators and digital wristwatches. Cell phones must be turned off during the examination.

Assignment submission policy:

All assignments are to be submitted in class. Any assignment submitted past the due date will incur a penalty of 50%. No assignments will be accepted one week past the due date.

Missed or Delayed Term Work:

You are expected to complete all assignments and hand them in on time, and you are expected to write the exams at the scheduled time. If you don't hand in some term work due to illness or another legitimate reason, Michael Jack is willing to make accommodations, but only if you report the problem to Michael Jack (not anyone else) as soon as is reasonably possible. Accommodations for missed term work will be decided on a case-by-case basis; typically they will involve either accepting assignments after the due date or making changes to the weighting used to compute the course grade.

Academic Dishonesty:

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure in an assignment to suspension from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's Academic Dishonesty Policy which is printed in the University Calendar.

Get help, but do not cheat!

You are encouraged to discuss the assignments with the instructor, lab adviser, and fellow students, since this is one of the best ways to learn the material. However, you should not let anyone solve the exercises for you. When you hand in your assignments, ask yourself two questions:

*Do I understand all the material I am handing in?
Could I do this assignment over again without any help?*

The answer to both questions should be YES. Computers allow electronic copying of programs and documents, which makes it very easy to cheat in a course like COSC 232. If you are caught cheating you will be reported to the Dean's Office for appropriate discipline. If you cheat and don't get caught, you are still in trouble, because examination marks count more than assignments marks in your final course grade.

Access to Instruction:

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Disability Services Office (BL Suite 109, 748-1281, disabilityservices@trentu.ca) as soon as possible. Complete text can be found under Access to Instruction in the Academic Calendar.

May 4, 2008

To whom it may concern:

Letter of Recommendation for Mr. Michael Jack

It is a rare occasion when one has the opportunity to support, without equivocation, the application of someone who fully demonstrated the qualities of industry, perseverance, maturity, discipline, and scholastic ability. Michael Jack meets and exceeds these criteria.

When Michael arrived at Trent University in January 2001, he plunged into his studies with extraordinary enthusiasm and dedication. He literally absorbed course material, seeking to understand every minutiae of each subject. Consequently, Michael's academic performance in computer science and mathematics was outstanding. With an overall average above 90%, Michael earned a prestigious postgraduate scholarship from the Natural Sciences and Engineering Research Council of Canada which he held throughout his two-year tenure as a graduate student at Trent University. Two accomplishments at Trent, however, bear further recognition.

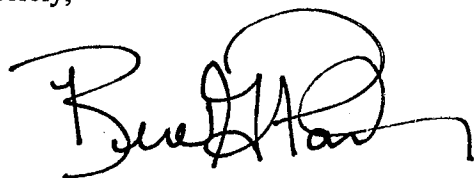
1. At the end of only his second year of undergraduate study at Trent, I engaged Michael as a Research Assistant. For four months during the summer of 2002, Michael read and critiqued scholarly papers in parallel job scheduling. In addition, he developed simulation models based on the paradigm of marginal utility, culminating in the publication of two conference papers in 2002 and 2003. It was this interest in simulation and modeling that attracted Michael to the graduate program at Trent University where he completed, under my supervision, his master's degree in 2006.
2. In his final year of undergraduate study, Michael was also part of a software engineering team that was entrusted to design and implement a scheduling/booking system for the Department of Athletics at Trent University. With his drive, natural leadership abilities, and excellent communication skills, the project met its deadlines, satisfied its requirements, and was, in my view, the most well-rounded and well-prepared final-year project since my arrival at Trent in 2001.

Michael is by far the hardest working student that I have had the opportunity to watch firsthand. And his work ethics extend to all facets of his life. Everything Michael sets out to do is tackled to the very best of his abilities. Nothing is left undone or left to chance. He views obstacles as opportunities to improve himself. He sets very high standards and pursues life with integrity and unabashed honesty. He is able to size up situations objectively and see through the masks of others.

Michael also brings to the table a unique wealth of experience. As a native of Russia, Michael immigrated with his family to Israel where he worked and studied with the Israeli navy. He has also traveled to ports of call around the globe as a merchant marine. Michael then moved to the United States in the late 1990's as an employee of Intel Corporation. There, he perfected his English and is now fully trilingual in Russian, Hebrew, and English. His written and oral writing abilities in English are superb and indeed, exceed most native speakers of the language.

Michael is well-read and extremely fit. He is outgoing and generous. And over the years, I have developed a great deal of respect for Michael and his life goals. He would be a wonderful asset for any organization and I wish him every success.

Sincerely,



Brian G. Patrick, PhD MBA
Chair and Associate Professor
Computing & Information Systems
Trent University

COMPUTER SCIENCE/STUDIES PROGRAM

1600 West Bank Drive
Peterborough, ON Canada K9J 7B8

Telephone (705) 748-1011 x 1495
Facsimile (705) 748-1066
E-mail compstudies@trentu.ca
Web <http://www.trentu.ca/csp>

May 6, 2008

Reference for Michael Jack

I have had the pleasure of knowing Michael Jack since the fall of 2001. In that time I have had dealings with him on three levels: as an undergraduate student, as a graduate teaching assistant and finally as a teaching colleague.

As an undergraduate student, Michael would rank in the top 3 students out of the hundreds that I have taught over the last 10 years. He was always more than prepared and always did more than what was asked of him. On the rare occasion when he had a problem understanding a concept he would come to me for clarification, which illustrates his desire to learn a topic completely. In his fourth-year software engineering project course, he assumed a team leadership role and led his team to a very successful conclusion.

Michael's duties as my graduate teaching assistant covered two courses. In my third year requirements course, he led weekly seminars and for the fourth year project course he assisted students with their projects. I was always confident that any task that I gave Michael would be well done and I never had to give it a second thought. In the seminars, I would tell Michael the articles that he should read and then he would lead classroom discussions on the topics. He showed a great skill in being able to read between the lines in the articles and then to be able to pass this knowledge along to the students.

As a teacher, Michael has proven that he can develop courses independently and that he has excellent time management and organizational skills. He has taken it upon himself to rewrite existing course notes to his standards and to try and develop courses in which the students have the best opportunity to succeed. He has proven that he can deal with students in a fair and ethical manner.

In conclusion, I highly recommend Michael Jack. He has shown me that he has the desire, dedication, and ability to succeed and I would trust him with any task given him. I am positive that Michael will be successful in any future endeavour.



Brian Hircock, Lecturer
Office phone 705 748-1011 ext 7768
Home phone 705 295-1536
Email: bhircock@trentu.ca



TRENT UNIVERSITY

Department of Computing and Information Systems
1600 West Bank Drive, Peterborough, Ontario, Canada K9J 7B8
Tel: (705) 748-1011 x7542, Fax: (705) 758-1066
Email: rhurley@trentu.ca

May 8, 2008

To Whom it May Concern,

I am very pleased to be providing a letter of reference for Michael Jack who completed a M.Sc. degree in the Applications of Modeling Program and an B.Sc. degree in Computer Science/Studies both at Trent University. Since completion of his graduate degree, Michael has lectured on a part-time basis for the Program.

As an instructor, I had the great pleasure in having Michael in many courses (both undergraduate and graduate) and I have no difficulty in stating that he was the best student I had seen in many years (his marks in all of my courses were in the 90s). It was not just his academic ability which set Michael apart from other high-achieving students, it was his work ethics. He went above and beyond the course requirements and produced work that far exceeded expectations. Another area which distinguished Michael was that it was quite apparent that he truly wanted to learn new material; he had a thirst for knowledge.

As the Chair of the Program from 1999 to 2007, it was my responsibility to oversee the selection of part-time lecturers. As a graduate student, Michael had served as a guest lecturer in several of my course on numerous occasions. His maturity, clarity, and professionalism was well received by the students. As a result, the Program had no reservations in allocating courses for Michael to teach. He has taught our first-year software engineering course on two occasions, a second-year course in digital logic and a second year course in computer architecture. The feedback from the students has been positive and the Program would not hesitate to assign him future courses if he so desires. One of the strongest comments made by Michael's students is that he went out of his way to provide support in and out of the classroom. Michael had a real desire to see his students succeed at the same time ensuring that they received a quality education. Unfortunately, Michael's association with Trent is limited to a part-time basis as he does not have a Ph.D. which is a requirement for a full-time position.

As a undergraduate student, a graduate student, and now a part-time lecturer, Michael is a tremendous asset to our Department. I have no doubt he will excel in whatever he chooses to do and I am pleased to provide my highest recommendation to Michael Jack.

Sincerely,

Dr. Richard Hurley

Full Professor

Department of Computing and Information Systems
Trent University

May 8, 2008

To Whom It May Concern,

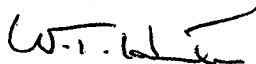
I first met Michael Jack seven years ago when he was a student at Trent University and selling door-to-door for Direct Energy during the summer. I used to be a professor at Trent University and still use the gym there for workouts and he was doing the same.

While I never taught Michael I followed his academic career with interest as he compiled an exceptional record in his courses. He won the prize for the highest mark in first year economics which was the discipline that I taught, although he majored in computer science.

My wife and I live on a farm and he has visited us on many occasions, including at least three times at Christmas. He has helped us by staying at the farm when we have been away, and has hunted on our property, successfully getting a deer with a traditional bow and arrow. When his family has visited from Israel we have entertained them here and we have been the guests at Michael's place on several occasions.

This is the background to our friendship with Michael and I can only say how impressed we are with his accomplishments. You will know that his academic record is exemplary. He has worked diligently at improving his English and his written skills now surpass most native-speaking students of my acquaintance. He has an enquiring mind and reads diligently in a number of fields and is anxious to discuss the issues raised. He is highly disciplined and was extremely focused on his studies, and has continued to apply the same energy to his teaching duties. With respect to his hobby of guns he is fastidious about knowing the rules and following them. And of course he is extremely fit.

It has been a great pleasure for us to get to know Michael and we would certainly recommend him most highly for any position for which he might apply.



William T. Hunter,

R.R.# 8,
Peterborough,
Ontario K9J 6X9

11-May-08

To Whom It May Concern,

Letter of Reference for Michael Jack

Michael has demonstrated his strengths, values and character to me on many different levels over the past six years. His relationship to me has evolved over time from student and employee to a loyal friend.

Outstanding Leadership and Pride in Work:

I first met Michael when I moved to the Peterborough area from Toronto in 2002. My occupations since then have been running a software company on a full time basis and lecturing at Trent University on a part time basis. Michael happened to be in the first class I taught at Trent in the fall of 2002. Michael was clearly the leader of his peers in that class. Everyone, including myself, would look up to Michael when complex issues were identified in class. Michael, as the rest of the class, knew it was the first course I had ever taught and he helped me understand the University system and how to better deliver my course materials. In the following semester Michael was in a course I created for the University and he again showed leadership in class as well as outstanding pride in his work. His assignments and projects would be in the order of 200 pages while those of his peers were 20 pages. The qualities of his projects have yet to be matched, 6 years later.

Since then Michael has acted as teaching assistant for my courses at Trent as well as he helped me with a few projects at my software company. Michael has helped me in times I have gotten myself into work overload situations, where the work seemed unmanageable. He comes in, always calmly and pleasantly, in times of my crises, and helps me finish work in time frames that are almost inhuman.

Trusted and Loyal:

Michael quickly earned my trust, which was/is difficult to do. We had him house-sit and pet-sit many times. We would not trust just anyone in our house or with our pets. I remember the first time Michael came over to our house to deliver a project (we were on 5 acres at the time) and our dogs were not friendly to strangers. At the time I thought Michael would be arriving, I went outside to gather the dogs, a Rottweiler and a Staffitire Bull Terrier. Much to my surprise I found Michael and the two dogs playing and rolling on the ground like puppy mates. Dogs are very good judges of character. At least, these two were. I had never seen them accept a person as they did to Michael.

In closing, I would recommend Michael highly, as he takes himself and his work very seriously. His work ethic is unparalleled, and his dedication, pride and loyalty are unwavering. Michael is a very loyal and caring person. He is someone who can always be counted on.

Sincerely,



Sofie Andreou, M.Eng.
VP Community Development
The Breken Group
705-741-5864 x103
sofie@ylm.ca

12-May-08

Dear Michael,

On behalf of Trent University and the Department of Computing & Information Systems, I would like to extend my thanks to you for your recent contributions to Trent's undergraduate programs in your capacity as a CUPE course instructor.

I recognize what an important role CUPE course instructors play in delivering the high quality undergraduate instruction on which Trent prides itself. Your knowledge and skills are truly valued.

I hope you have enjoyed your experiences in the classroom, and I wish you well for the summer.

Sincerely,

Christine McKinnon
Dean of Arts & Science

To Whom It May Concern:

As a member of the Trent University athletics complex for the past seven years, and as an employee for the last three years I have come to know several members and students. The fall of 2002 marks the time to which I had the pleasure of meeting Michael Jack. To this day, it is without question that I have not yet met one individual who has demonstrated his level of dedication to physical health, education, family and friends. I believe the personal character which Michael displays is certainly unique, (mostly due to his level of dedication) he naturally embodies the idealizations to which most may only aspire to become.

Consistently over the past six years I have admired Michael's highly regimented schedule. He is absolutely incomparable in his level of efficiency; I would often find him exercising while also reading a textbook, typically regarding this time as his "break" between twelve hour days of work, reading and writing. Regardless to how busy Michael may be, he always reserves some time to teach other members on proper exercise, and engage in the occasional scholarly debate.

Without reservation I highly recommend Michael for this position, and truly believe he deserves the opportunity to demonstrate what I consider his unmatched level of qualification. Please feel free to contact me if you should have any further questions regarding this letter. Cody Harding-Calder

May 12/2008



14-May-08

To Whom It May Concern,

I first met Michael seven years ago when he came to our house as a sales representative for Direct Energy. We had a basement available for rent at the time and upon learning that Michael was a student at Trent University my wife inquired if he knew any students who were looking for accommodations. After Michael had checked the place out, he decided to move in himself and he has lived in our house ever since.

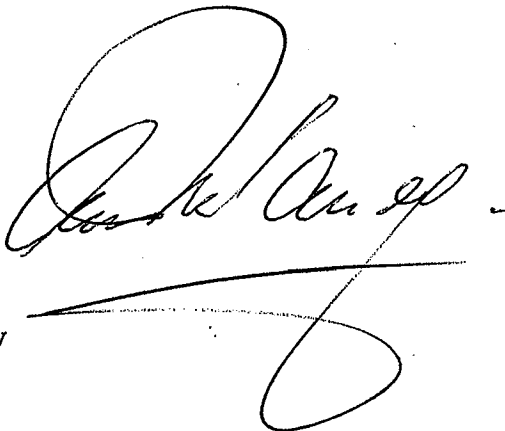
We have five children of our own. They are fully grown and have lives of their own. Over the years Michael has become part of our family. He is like a son to me. In November 2004 my wife had a brain aneurysm and she has been on disability since then. I am retired and drive school bus on a part time basis as I have to look after my wife.

Michael is extremely clean, organized, and diligent. I'd say "immaculate" would be the right choice of a word to describe his living environment. He is dedicated to anything and everything he sets out to do and has a positive attitude to everyone he meets. He has never caused any problems, never defaulted on his rent payments. Whenever I ask him to help me, he is right there right away. Over the years he has been helping us to maintain the property and he has personally renovated half of the house.

Michael's entire family has visited and stayed in our house in August 2003 and his brother has come to stay with us over the summers in 2006 and 2007.

Michael is loyal and trustworthy. I would without hesitation recommend him for any position for which he might apply.

Sincerely,

A handwritten signature in black ink, appearing to read "Andre Melaney", with a long horizontal flourish extending to the right.

Andre Melaney
705-652-8125



THE ONTARIO POLICE COLLEGE WITH APPROVAL OF
THE MINISTER OF COMMUNITY SAFETY AND CORRECTIONAL SERVICES,

PROVINCE OF ONTARIO

HAS CONFERRED UPON

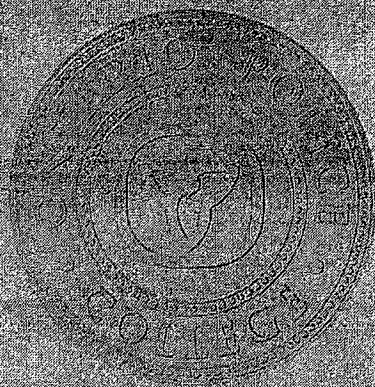
Michael JACK

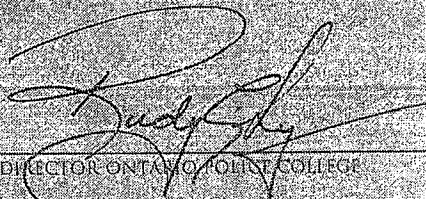
THE ONTARIO POLICE TRAINING SYSTEM DIPLOMA

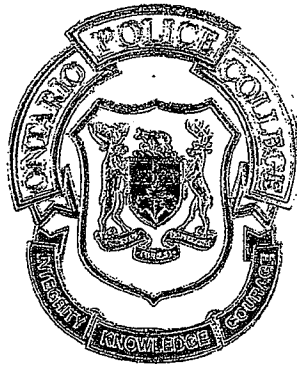
ATTESTING TO SUCCESSFUL COMPLETION OF THE

BASIC CONSTABLE TRAINING PROGRAM

November 27, 2008




DIRECTOR, ONTARIO POLICE COLLEGE



THE ONTARIO POLICE COLLEGE WITH APPROVAL OF
THE MINISTER OF COMMUNITY SAFETY AND CORRECTIONAL SERVICES,
PROVINCE OF ONTARIO,
HAS CONFERRED UPON

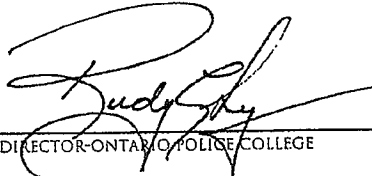
Michael JACK

THE ONTARIO POLICE TRAINING SYSTEM DIPLOMA
ATTESTING TO SUCCESSFUL COMPLETION OF THE

BASIC CONSTABLE TRAINING PROGRAM

November 27, 2008




DIRECTOR-ONTARIO POLICE COLLEGE



ONTARIO POLICE COLLEGE
Ministry of Community Safety and Correctional Services
BASIC CONSTABLE TRAINING PROGRAM
Student Evaluation

Name JACK, Michael
Service Ontario Provincial Police

Date Sep 3 – Nov 27, 2008
Class BC-0828

The wide variety of subject areas in Basic Constable Training have been grouped into various categories (see attached BCT Program specifications). Certain aspects of each subject area that require pen and paper testing are assessed in midterm and final examinations, the overall pass mark being 75% in each of the seven examinations listed below.

Examination	Student Mark (%)	Course Average (%)
Community Policing (Closed Book) <i>Ethics / Community Policing / Anti-Racism / Domestic Violence / Death Notification / CPIC / Crime Scene Preservation</i>	Met Standard	
Federal Law (Closed Book) <i>Arrest / Criminal Offences / Evidence / Drugs / YCJA</i>	90.2	81.1
Traffic Law / Federal Law (Open Book) <i>Arrest / Criminal Offences / Drugs / YCJA / Highway Traffic Act / Criminal Driving Offences</i>	86.0	84.1
Police Vehicle Operations (Closed Book) <i>Legal Responsibilities / Vehicle Dynamics / Environmental Factors / Suspect Apprehension Pursuits Regulation</i>	95.0	86.8
Provincial Law (Closed Book) <i>POA / MHA / LLA / Child and Family Services Act / Children's Law Reform Act / Family Law Act / Trespass to Property Act / HTA / Police Services Act</i>	90.3	82.2
Use of Force/Officer Safety (Closed Book) <i>Defensive Tactics / Firearms / Officer Safety / Tactical Communications / Use of Force / Study Guide / Use of Force Model</i>	96.7	89.1
Average	91.6	81.5

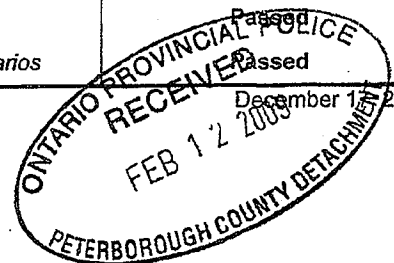
Physical Skills Training	
Defensive Tactics	Skill Components / Applied Scenarios
Firearms	Skill Components / Applied Scenarios
Police Physical Training	PREP
Police Vehicle Operations	Skill Components / Applied Scenarios
Retested Police Vehicle Operations	

Passed

Passed

Passed

Passed



December 17, 2008

... 2

JACK, Michael
Ontario Provincial Police
BC-0828

General Comments

Comments pertaining to training issues are generally reserved for those students who either demonstrated outstanding capabilities or experienced significant difficulties in various training activities. Lack of written comments in the training areas listed below indicates that the candidate performed satisfactorily in these components and progressed steadily throughout the course.

Defensive Tactics

This student has successfully completed an Aerosol Weapon exam to comply with Police Services Act Regulations and Policing Standards guidelines.

Firearms

Police Physical Training

All Basic Constable Training students are given the opportunity to test for and receive the Ontario Police Fitness Award. This candidate is to be commended for receiving a perfect score of 100% on this fitness test.

The Physical Training Department recognizes this student for demonstrating a high proficiency level in both the Ontario Police Fitness Award and the PREP test.

Police Vehicle Operations

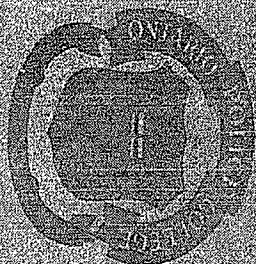
Special Mention

Comments

Diploma Awarded.

EXHIBIT 12

ONTARIO POLICE COLLEGE



Congratulations

Let It Be Known That

Michael Jack

Received

**100 % on the Ontario
Police Fitness Award**

During Basic Constable Training

Chris Hara

Physical Supervisor

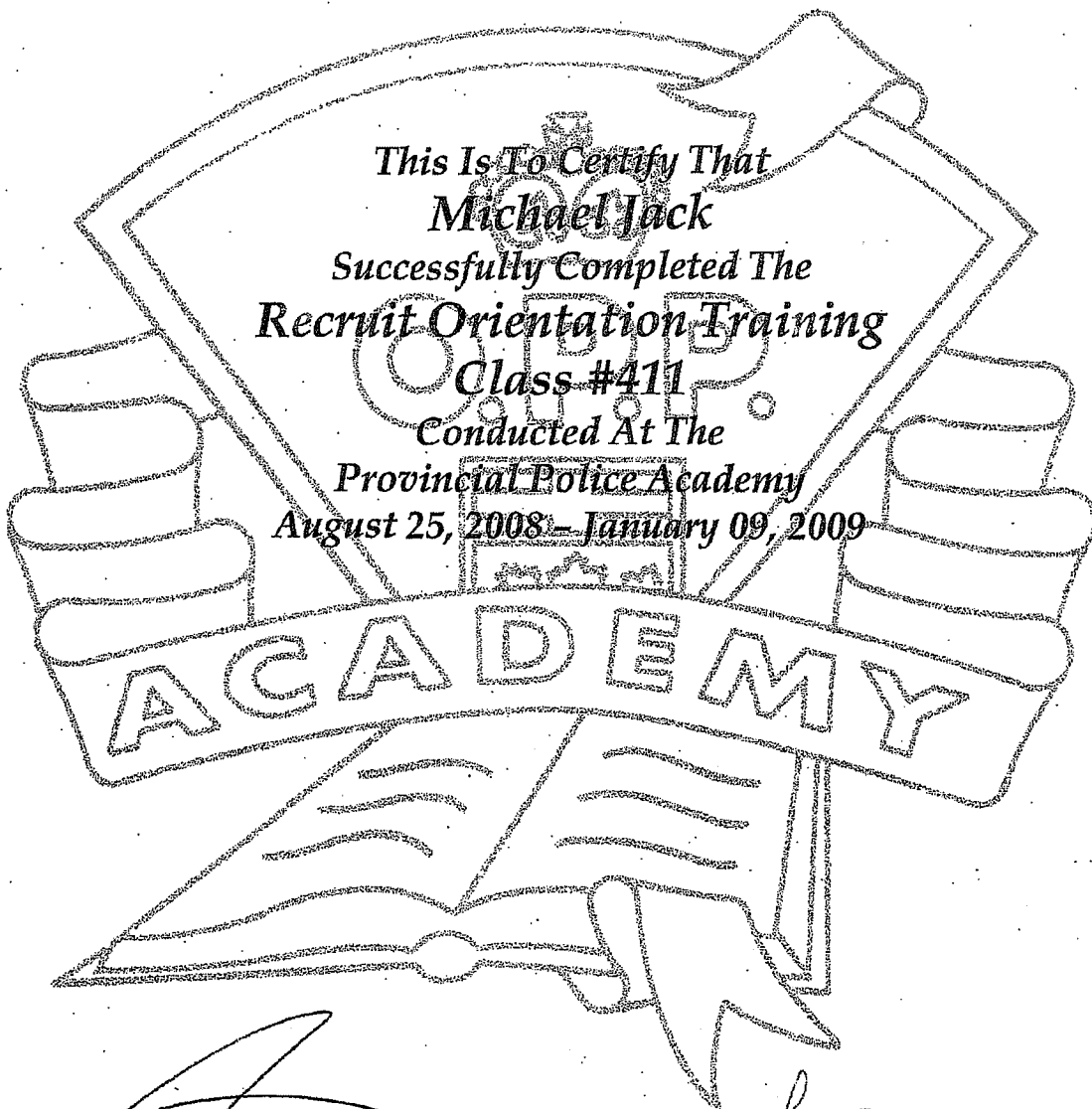
Doni Shields
Physical Staff

*Dated at Rymer
this day of November 27, 2008*



Ontario Provincial Police

*This Is To Certify That
Michael Jack
Successfully Completed The
Recruit Orientation Training
Class #411
Conducted At The
Provincial Police Academy
August 25, 2008 – January 09, 2009*



Director

Course Co-ordinator



Trent University
COIS-ADMN 1620H
2010 Summer Term, Peterborough

INTRO TO INFORMATION SYSTEMS

Course Outline

Instructor: Michael Jack,
OC 102.11
(705) 748-1011 ext. 7251
michaeljack@trentu.ca

Secretary: Bonnie MacKinnon
OC 102.6
(705) 748-1011 ext. 7802
cois@trentu.ca

Text: MIS 2010,
H. Bidgoli (2010),
Course Technology, Cengage Learning (www.cengage.com)

Meetings: Monday/Wednesday, 18:30 - 21:30, SC 115.

Office Hours: Monday/Wednesday, 17:30 - 18:30, OC 102.11.

Course Content:

This course provides an introductory overview of information systems and will address the following important questions:

- Information Systems: An Overview
- Computers: The Machines behind Computing
- Database Systems, Data Warehouses, and Data Marts
- Personal, Legal, Ethical, and Organizational Issues of Information Systems
- Protecting Information Resources
- Data Communication: Delivering Information Anywhere and Anytime
- The Internet, Intranets, and Extranets
- E-Commerce
- Global Information Systems
- Building Successful Information Systems
- Enterprise Systems
- Management Support Systems
- Intelligent Information Systems
- Emerging trends, Technologies, and Applications

Course Objectives:

- Understand why information systems matter to contemporary organizations and society.
- Understand exactly what innovative organizations are doing with contemporary information systems and where things are headed.
- Empower students with the essential knowledge they need to be successful in the use and understanding of information technology in their careers.
- Teach the students to use myLearningSystem (WebCT) system.

Methodology:

For each student, this course is comprised of weekly lectures, nine quizzes, two tests, and a research project. It is a busy agenda, but one that attempts to involve you as much as possible in the learning process.

Grading scheme:

50%	6 online quizzes
50%	2 tests
5%	bonus

Quizzes:

The quizzes are based on the chapters from the text book – one quiz per two chapters. The best 5 out of 6 will be selected for the calculation of the final grade. All the quizzes will be done online using the myLearningSystem (WebCT) system. Since you are taking quizzes at the time and place of your convenience (during the allocated time frame), you can use any resources such as the Internet and books to help you answer the questions. However, the quizzes must be completed independently. The duration of each quiz is 20 minutes.

Tests:

Both tests are paper based and will be given in class settings. The first test covers course text book chapters 1 – 6 exclusively. The second test covers course text book chapters 7 – 12 exclusively. The tests are open text book. The duration of each test is one hour. The tests are structured strictly around material presented in the course text book chapters. The better you do on your quizzes, the easier it will be for you to score high on the tests.

No computers or calculators may be used during the tests. The only digital devices permitted are digital wristwatches. Cell phones must be turned off as well.

Bonus:

You can earn up to 5% bonus that will be added to your final grade. For every logical or syntax mistake that you find in the text book you can earn a point. The point will be awarded to ONLY the first student who shows the instructor the mistake and for a maximum of 5 points per student.

Missed or Delayed Term Work:

You are expected to take the online quizzes within the allocated time frame. If you miss taking the quiz due to illness or another legitimate reason, the instructor is willing to make accommodations, but only if you report the problem to the instructor (not to anyone else) as soon as reasonably possible. Accommodations for missed term work will be decided on a case-by-case basis. Typically they will involve either taking the quiz after the expiration date or making changes to the weighting used to compute the course grade.

Correspondence:

Due to privacy legislation, any email correspondence from the course instructor or assistant will be forwarded to your Trent account only.

Academic Dishonesty:

Academic dishonesty, which includes plagiarism and cheating, is an extremely serious academic offence and carries penalties varying from failure in an assignment to suspension from the University. Definitions, penalties, and procedures for dealing with plagiarism and cheating are set out in Trent University's Academic Dishonesty Policy which is printed in the University Calendar and found at:

http://www.trentu.ca/deansoffice/policies_dishonesty.php

Get help, but do not cheat!

You are encouraged to discuss your work with the instructor, lab adviser, and fellow students, since this is one of the best ways to learn the material. However, you should not let anyone take your quizzes. Before you take a quiz ask yourself two questions:

*Do I understand the relevant text book material?
Could I do this over again without any help?*

The answer to both questions should be YES. If you are caught cheating you will be reported to the Dean's Office for appropriate discipline.

Access to Instruction:

It is Trent University's intent to create an inclusive learning environment. If a student has a disability and/or health consideration and feels that he/she may need accommodations to succeed in this course, the student should contact the Disability Services Office (BL Suite 109, 748-1281, disabilityservices@trentu.ca) as soon as possible. Complete text can be found under Access to Instruction in the Academic Calendar.

Course Schedule

Lecture Dates	Material
June 21	Ch 1: Information Systems: An Overview.
June 23	Ch 2: Computers: The Machines behind Computing.
June 28	Ch 3: Database Systems, Data Warehouses, and Data Marts.
June 30	Ch 4: Personal, Legal, Ethical, and Organizational Issues of Information Systems.
July 5	Ch 5: Protecting Information Resources.
July 7	Ch 6: Data Communication: Delivering Information Anywhere and Anytime.
July 12	Test 1 and Ch 7: The Internet, Intranets, and Extranets.
July 14	Ch 8: E-Commerce.
July 19	Ch 9: Global Information Systems.
July 21	Ch 10: Building Successful Information Systems.
July 26	Ch 11: Enterprise Systems.
July 28	Ch 12: Management Support Systems.
July 30	Test 2

Quiz #	Timeframe		Material
	Release	Close	
1	June 23, 9:00 pm	June 28, 2:00 pm	Chapters 1 & 2
2	June 30, 9:00 pm	July 5, 2:00 pm	Chapters 3 & 4
3	July 7, 9:00 pm	July 12, 2:00 pm	Chapters 5 & 6
4	July 14, 9:00 pm	July 19, 2:00 pm	Chapters 7 & 8
5	July 21, 9:00 pm	July 26, 2:00 pm	Chapters 9 & 10
6	July 28, 9:00 pm	August 2, 2:00 pm	Chapters 11 & 12

Final date to drop the course without academic penalty is Thursday, July 15th, 2010.